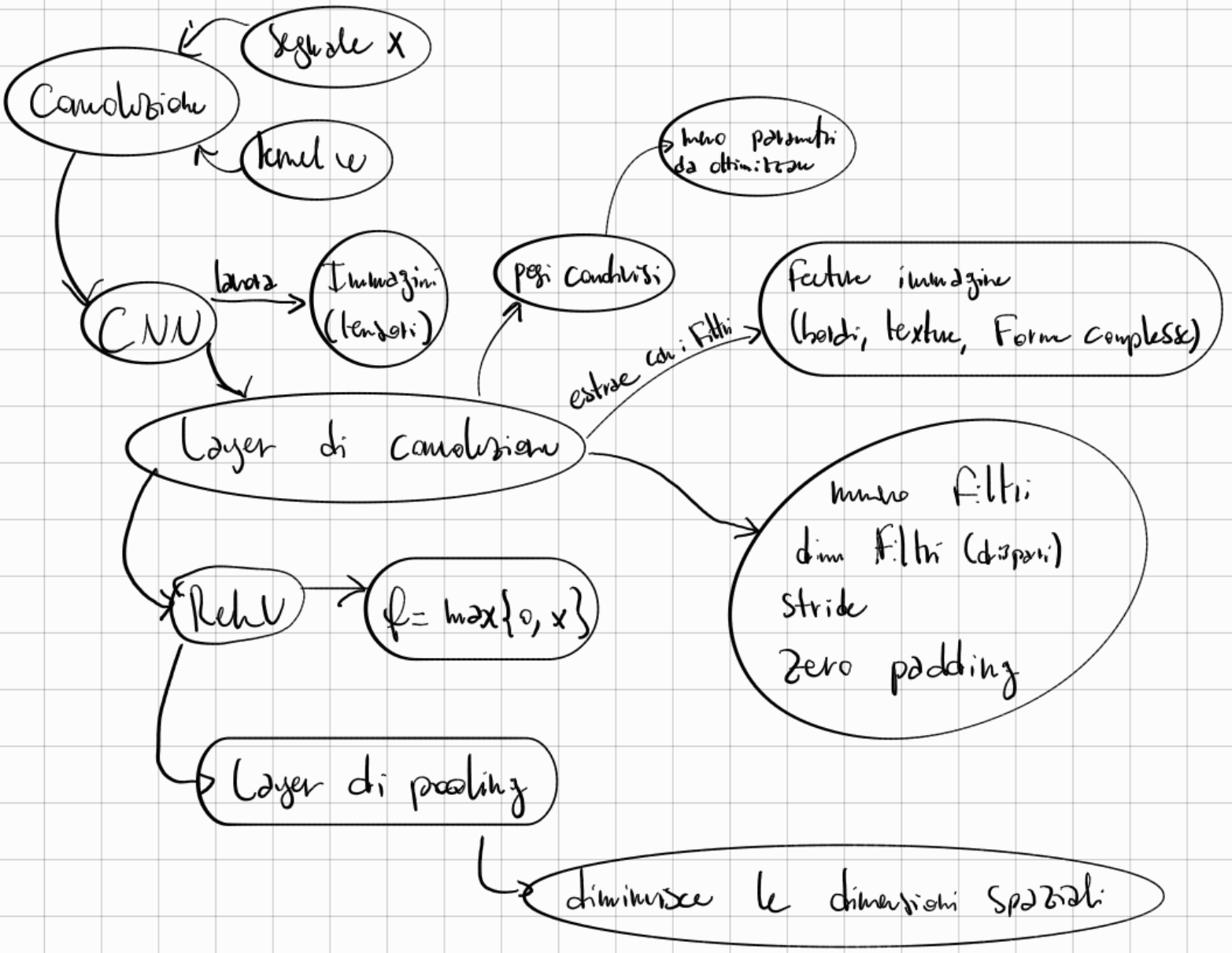
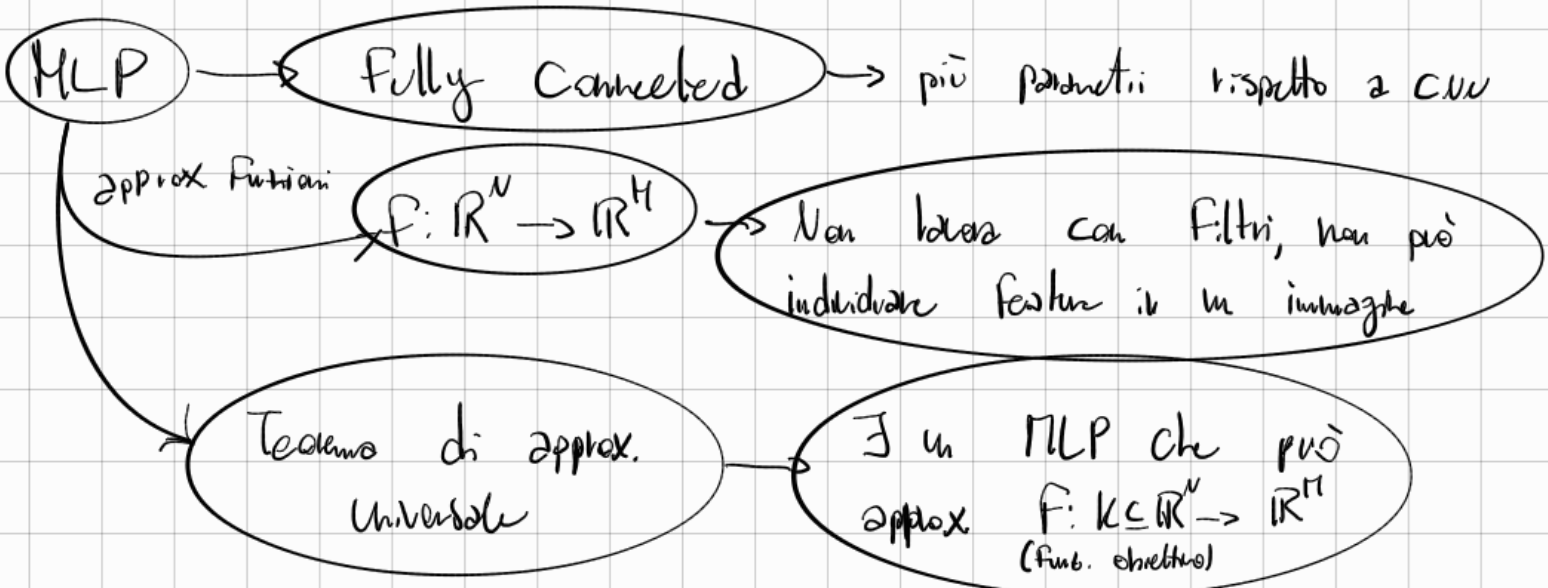
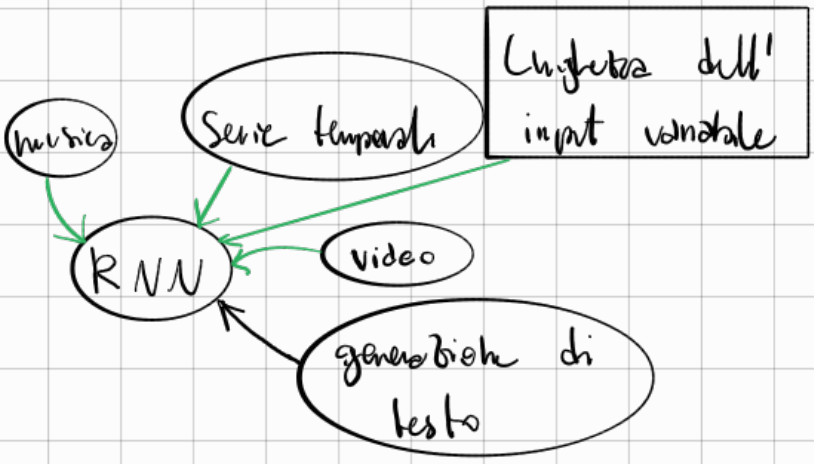
Operazione convoluzione: $(x * w) = \int_{\mathbb{R}} x(a)\, w(t-a)\, da$ $N=1$

Proprietà: commutativa, lineare

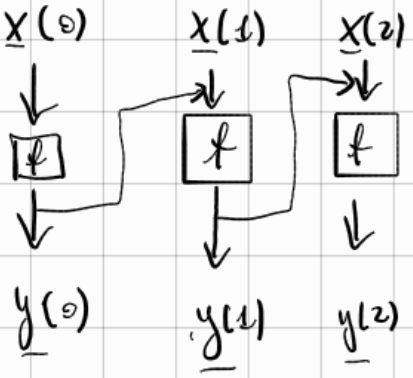
invariante rispetto al tempo ($N=1$) e allo spazio ($N=2$) $(X * W)(u,v) = \int_{\mathbb{R}^2} X(a,b)\, W(u-a, v-b)\, da\, db$

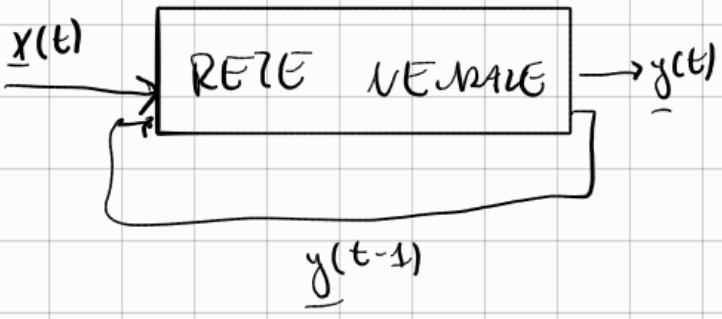





Architettura RNN:




$$\sigma\left(W \cdot \underline{x}(t) + \hat{W} \cdot \underline{y}(t-1) + \underline{b}\right)$$

Segnale corrente — Segnale prec.